PARTICLE TRACK RECONSTRUCTION AS A SPIN-GLASS OPTIMIZATION PROBLEM

Giovanni Guidarini & Nicola Mucciaccio
Report by Giovanni Guidarini
gguidarini@student.ethz.ch
Legi Number: 25-945-684

ABSTRACT

Track reconstruction in high-energy physics (e.g. at CERN) is a large combinatorial problem. This project maps a simplified 2D version of the task onto an Ising-like spin-glass model. Potential track segments between detector layers are treated as binary variables (spins). By designing a Hamiltonian that rewards smooth trajectories and strongly penalizes bifurcations, track finding becomes an energy minimization problem solved via simulated annealing. This report presents our objectives, the methodology and the shortcomings of our results.

1 INTRODUCTION

At CERN, the Large Hadron Collider (LHC) produces billions of particle collisions per second; when a collision occurs, a spray of charged particles is emitted from the collision point. The resulting particle trajectories are recorded by detectors which consist of multiple layers of sensors that capture the passage of charged particles as discrete hits. Reconstructing the original particle tracks from these hits is a critical step in analyzing the collision data [2]. This task is computationally challenging due to the sheer volume of data and the combinatorial nature of the problem, as multiple hits can be combined in various ways to form potential tracks.

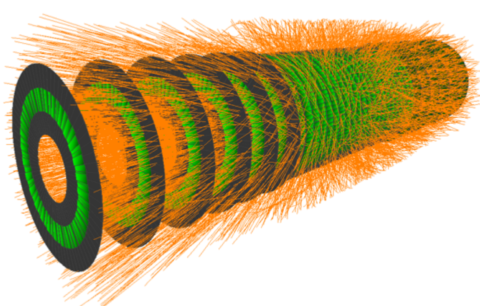

Figure 1: An event display of trajectories in the OpenDataDetector, a simulated detector similar to the ones used in high-energy physics experiments. Credit: CERN

The goal of this project is to present a physics-inspired approach to track reconstruction by formulating it as an optimization problem following the approach pioneered by Denby [3] and Peterson [7]. We model the potential track segments as binary variables (spins) in an Ising-like spin-glass model. By designing a Hamiltonian that encodes the physical constraints of track formation, we can use the Metropolis-Hastings algorithm [6] to find low-energy configurations that correspond to plausible particle tracks. An important simplification in our model is that we consider a 2D projection of the detector, which allows us to focus on the core combinatorial aspects of track reconstruction while avoiding the complexities of 3D geometry. We work with a synthetic dataset that simulates the hits produced by particle collisions, and we evaluate our method based on its ability to correctly identify the original tracks. The simulation kernel is implemented in C++ for performance reasons, while the analysis and visualization of results are done in Python.

2 Synthetic Detector Model

In order to generate synthetic data for our track reconstruction problem, we simulate a simplified 2D detector model. We assume that all particles originate from the same point (the collision vertex) and move outward following circular trajectories due to a uniform magnetic field perpendicular to the plane of motion. The detector consists of multiple concentric layers of sensors, centered around the collision vertex. When a particle passes through a layer, it leaves a hit at the point of intersection.

For our simulations, we assumed that the particles are scattered uniformly in angle and have a uniform distribution of transverse momenta, which determines the curvature of their trajectories. These assumptions are embedded in the track generation process: we randomly sample an angle ϕ, a radius of curvature r, related to the transverse momentum through the Lorentz force, and a direction of motion (clockwise or counterclockwise). The hits are then generated by calculating the intersection points of the circular trajectories with the detector layers.

In our experiments, we used a detector with 5 layers, positioned at radii of 1, 2, 3, 4, and 5 units from the collision vertex. Each collision produces 20 particles, resulting in a total of 100 hits across the layers. After generating the raw hits, we apply a small amount of Gaussian noise to the hit coordinates to simulate measurement uncertainties, and add a few random noise hits, distributed as a Poisson process with a mean of 2 noise hits per layer, to mimic the presence of spurious signals in the detector. One example of a simulated event is shown in Figure 2.

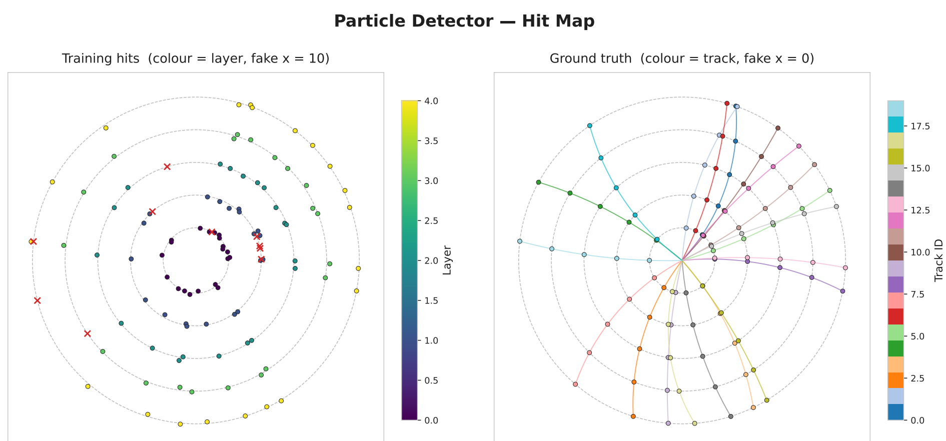

Figure 2: Example of a simulated event with 20 particles and 5 detector layers. The left plot shows the noisy training hits, with the real hits colored according by their layer and the fake hits as red crosses. The right plot shows the original tracks, with the hits they produce on each detector layer.

3 Ising Formulation of Segment Selection

The core idea of our approach is to represent track reconstruction as an optimization problem over binary variables, following the general line of the Hopfield-network formulations of [3; 7]. We first build the set of all candidate segments connecting hits in adjacent detector layers. Each candidate segment i is associated with a binary variable $x_i \in \{0, 1\}$, where $x_i = 1$ means that the segment is selected and $x_i = 0$ means that it is not.

The objective implemented in our code has three contributions:

1. A pairwise interaction term, encoded by a sparse matrix $\boldsymbol{J}$, that rewards compatible concatenations of segments and penalizes incompatible ones such as forks, merges, and sharp turns.

2. A local field term, encoded by a vector $\boldsymbol{h}$, that penalizes long segments and, for segments between the first two detector layers, penalizes directions that deviate too much from the local radial direction.
3. A three-segment interaction term that rewards triplets of consecutive segments whose local turning angles are mutually consistent, and penalizes triplets that are not.

The Hamiltonian minimized by the simulated annealing code is

$$\mathcal{H}(\boldsymbol{x}) = -\frac{1}{2}\sum_{i,j} J_{ij}x_ix_j - \sum_i h_ix_i + \sum_{(j,i,k)\in\tau} E_{\text{triplet}}(j,i,k)\, x_jx_ix_k, \tag{1}$$

where $x_i \in \{0,1\}$, J_{ij} are the pairwise interaction weights, h_i are the local-field coefficients, and τ is the set of ordered triplets of concatenated segments (j,i,k) such that segment j ends where i starts and segment i ends where k starts. Because of the overall minus signs in the pairwise and local-field terms, positive values of J_{ij} favor the simultaneous selection of segments i and j, whereas negative values of J_{ij} penalize it; similarly, negative values of h_i act as penalties on selecting segment i. The $\frac{1}{2}$ factor in front of the pairwise term is a standard convention to avoid double-counting pairs.

An important detail of the implementation is that curvature is encoded twice: first as an additive correction to some pairwise couplings J_{ij}, and second as an explicit three-body term E_{triplet}.

4 Hamiltonian Terms Construction

4.1 Interaction Matrix

The interaction stage of the pipeline constructs the candidate segments and the sparse pairwise interaction matrix $\boldsymbol{J}$. Given two detector layers l and $l+1$, the code creates *all* possible segments joining one hit in layer l to one hit in layer $l+1$. Therefore, every binary variable corresponds to one segment between adjacent layers.

For each pair of distinct segments (i,j), the code assigns a coupling J_{ij} according to the following rules.

Let

$$a_{ij} = \frac{\boldsymbol{s}_i \cdot \boldsymbol{s}_j}{\|\boldsymbol{s}_i\|\,\|\boldsymbol{s}_j\|}, \tag{2}$$

where $\boldsymbol{s}_i = (dx_i, dy_i)$ and $\boldsymbol{s}_j = (dx_j, dy_j)$ are the segment direction vectors. Thus, a_{ij} is the cosine of the angle between the two segments.

- If i and j are *concatenated* segments, namely one ends exactly where the other begins, then they form a candidate chain across three consecutive layers. In this case, the code first checks whether
$$a_{ij} > \cos(\theta_{\max}). \tag{3}$$
If this condition is not satisfied, the pair is penalized with
$$J_{ij} = -\lambda_\theta, \tag{4}$$
where $\lambda_\theta > 0$ is the angle penalty.
- If the concatenated pair satisfies $a_{ij} > \cos(\theta_{\max})$, the code starts from the alignment score a_{ij} and then adds a curvature-dependent correction. Denote by $\phi_i = \operatorname{atan2}(dy_i, dx_i)$ the absolute direction angle of segment i, and let $\Delta\phi_{ij}$ be the difference between the two segment angles, wrapped to the interval $[-\pi,\pi]$ after ordering the pair so that the earlier segment comes first in layer order. Then the implemented coupling is
$$J_{ij} = \begin{cases} a_{ij} + \beta_c, & \text{if } |\Delta\phi_{ij}| < \theta_{\text{tol}}, \\ a_{ij} - \lambda_c, & \text{otherwise}, \end{cases} \tag{5}$$
where $\beta_c > 0$ is the curvature bonus, $\lambda_c > 0$ is the curvature penalty, and θ_{tol} is the curvature tolerance.

- If the two segments end on the same hit, the pair is treated as a *merge* conflict and the code sets
$$J_{ij} = -\lambda_{\mathrm{m}}, \tag{6}$$
where $\lambda_{\mathrm{m}} > 0$ is the merge penalty.
- If the two segments start from the same hit, the pair is treated as a *fork* conflict and the code sets
$$J_{ij} = -\lambda_{\mathrm{f}}, \tag{7}$$
where $\lambda_{\mathrm{f}} > 0$ is the fork penalty.
- All other pairs receive no interaction, i.e. $J_{ij} = 0$.

With the sign convention of Equation 1, positive couplings reward compatible pairs, while negative couplings penalize incompatible ones.

4.2 Local Field Vector

The local field vector $\boldsymbol{h}$ is assembled by the annealing stage from the segment list. For each segment i, let ℓ_i be its Euclidean length. The code first applies a length penalty,
$$h_i \mathrel{-}= \lambda_\ell \, \ell_i, \tag{8}$$
where $\lambda_\ell > 0$ is the length-penalty coefficient.

An additional penalty is applied only to segments connecting layer 0 to layer 1. Let $\phi_i = \mathrm{atan2}(dy_i, dx_i)$ be the segment direction angle, and let
$$\rho_i = \mathrm{atan2}(y_{a_i}, x_{a_i}) \tag{9}$$
be the radial angle of the segment's starting hit in layer 0. The code computes the wrapped angular distance between ϕ_i and ρ_i; if this distance exceeds a threshold θ_{rad}, it applies the penalty
$$h_i \mathrel{-}= \lambda_{\mathrm{r}}, \tag{10}$$
where $\lambda_{\mathrm{r}} > 0$ is the layer-$0 \to 1$ radial penalty.

Therefore, the local field used by the code can be written as
$$h_i = -\lambda_\ell \, \ell_i - \mathbf{1}_{\{L_a(i)=0,\ L_b(i)=1\}} \mathbf{1}_{\{\Delta_{\text{rad}}(i) > \theta_{\text{rad}}\}} \lambda_{\mathrm{r}}, \tag{11}$$
where $\Delta_{\text{rad}}(i)$ is the wrapped angular distance between the segment direction and the local radial direction. Because the Hamiltonian contains the term $-\sum_i h_i x_i$, these negative values of h_i act as penalties on selecting long or non-radial segments.

4.3 Three-segment Interaction

The set τ of valid triplets consists of all ordered triplets of concatenated segments (j, i, k) such that segment j ends where segment i starts, and segment i ends where segment k starts. Since each segment already connects adjacent detector layers, these triplets span four consecutive layers.

For each $(j, i, k) \in \tau$, the code defines
$$E_{\text{triplet}}(j, i, k) = \begin{cases} -\beta_c, & \text{if } |\kappa(j,i) - \kappa(i,k)| < \theta_{\text{tol}}, \\ \lambda_c, & \text{otherwise}, \end{cases} \tag{12}$$
where
$$\kappa(j,i) = \mathrm{atan2}(dx_j \, dy_i - dy_j \, dx_i,\ dx_j \, dx_i + dy_j \, dy_i) \tag{13}$$
and
$$\kappa(i,k) = \mathrm{atan2}(dx_i \, dy_k - dy_i \, dx_k,\ dx_i \, dx_k + dy_i \, dy_k)\,. \tag{14}$$
Hence, the triplet term rewards chains whose two consecutive turning angles are similar, and penalizes chains whose turning angles are inconsistent with a smooth circular trajectory.

The same three hyperparameters β_c, λ_c, and θ_{tol} appear both in the pairwise curvature correction inside J_{ij} and in the explicit three-segment term above. This is exactly how the current implementation defines the energy function.

5 Simulated Annealing Algorithm

To find low-energy configurations of the Hamiltonian defined in Equation 1, we use the Metropolis-Hastings algorithm, which is a Markov Chain Monte Carlo (MCMC) method [6] that allows us to sample from the distribution defined by the Hamiltonian. The algorithm starts with a random initial configuration of the binary variables (spins), and iteratively proposes changes to the configuration by flipping the state of a randomly selected variable. The proposed change is accepted with a probability that depends on the change in energy $\Delta\mathcal{H}$ and a temperature parameter T:

$$\mathbb{P}_{\text{accept}} = \min\left(1, \exp\left(-\frac{\Delta\mathcal{H}}{T}\right)\right) \tag{15}$$

The temperature T is gradually decreased according to a cooling schedule, following the simulated annealing optimization technique [4]. This technique allows the algorithm to explore the energy landscape and escape local minima in the early stages, while converging to a low-energy configuration in the later stages. The cooling schedule of choice is a linear schedule, where the temperature ranges linearly from a maximum value T_{max} to a final value T_{min} over a fixed number of iterations N_{iter}:

$$T(n) = T_{\text{max}} - \frac{n}{N_{\text{iter}} - 1}(T_{\text{max}} - T_{\text{min}}) \tag{16}$$

The hyperparameters of the simulated annealing algorithm, T_{max}, T_{min}, and N_{iter} were optimized in a range that allows the algorithm to converge to a good solution within a reasonable amount of time. A Geometric cooling schedule was also implemented, but it did not yield better results than the linear one, so the results reported in this report are based on the linear cooling schedule. This does not necessarily mean that the linear schedule is superior to the geometric one, but in our case the linear one was a better fit without need for further tuning of the hyperparameters.

6 Experiments and Results

6.1 Hyperparameter Tuning

Before evaluating our method, we needed to tune the hyperparameters of our model, which include the penalty weights in the Hamiltonian and the parameters of the simulated annealing algorithm. We used a Bayesian optimization procedure to find the optimal values of these hyperparameters, with the objective of maximizing the efficiency of track reconstruction on an ensemble of 16 simulated events. We used the Optuna framework [1], which implements a Tree-structured Parzen Estimator for Bayesian optimization (for further details on Bayesian Optimization see [5], chapter 9). The optimization process was performed on The Euler cluster, using 48 to 64 workers in parallel; several iterations of the optimization process were required to find a good set of hyperparameters.

First, we aimed at optimizing the model for the noise-free scenario; this because if the model cannot perfectly reconstruct tracks in the absence of noise, it is unlikely to perform well in the presence of noise. After finding a good set of hyperparameters for the noise-free scenario, we then introduced noise in the hits and repeated the optimization process, starting from the hyperparameters found in the noise-free scenario.

6.2 Evaluation Metrics

The metrics we used to evaluate the performance of our method are the following:

- **TPR (Segment Recall)**: the fraction of true segments that are correctly identified by our method.
- **Precision (Segment Purity)**: the fraction of segments identified by our method that are actually true.
- **F1**: the harmonic mean of precision and recall, which provides a single metric that balances both aspects of performance:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{17}$$

- **Track Efficiency (Strict)**: the fraction of true tracks that are correctly reconstructed: a track is considered correctly reconstructed if all its segments are correctly identified by our method.
- **Track Efficiency (Soft)**: the fraction of true tracks that are almost correctly reconstructed: in our case, with 5 layers, there are 4 nontrivial segments in each track, so we consider a track to be almost correctly reconstructed if at least 3 out of 4 segments are correctly identified by our method.
- **Fake Rate**: the fraction of selected segments that are fake.
- **N° bifurcations**: the total number of hits where branching/merging appears in selected segments.

Additionally, we also considered some metrics relative to the optimization process, such as the final energy of the configuration found by the simulated annealing algorithm and the freeze temperature (i.e. the temperature at which c_{v} peaks).

To the quantitative metrics, we added a visual inspection of the reconstructed tracks, as this allows us to identify common failure modes of our method and to fix them by modifying the Hamiltonian. Indeed, some of the terms in the Hamiltonian (the three-segment interaction term and the local field term) were added and tuned after visually inspecting the reconstructed tracks and identifying common failure modes, such as the selection of segments that do not form smooth trajectories, and the selection of long segments that form unphysical tracks.

6.3 RESULTS

We evaluated our method on an ensemble of 128 simulated events, each containing 20 tracks and 5 layers. We report the performance of two models: one whose hyperparameters were optimized on noise-free events and one optimized on noisy events. Each model was tested on both noise-free and noisy test sets, yielding the four configurations summarized in Table 1.

Model (Opt. on)	Test Data	TPR	Precision	F1	Track Eff. (Strict)	Track Eff. (Soft)	Fake Rate
Noise-free	Noise-free	0.934	0.946	0.940	**0.853**	0.932	0.053
Noise-free	Noisy	0.295	0.465	0.361	**0.016**	0.102	0.535
Noisy	Noise-free	0.869	0.871	0.870	**0.631**	0.870	0.130
Noisy	Noisy	0.673	0.728	0.699	**0.307**	0.586	0.272

Table 1: Performance of both models (optimized on noise-free and noisy events, respectively) evaluated on noise-free and noisy test sets, averaged over 128 events.

In all the configurations, the number of bifurcations and merges is 0, which means that the merge and fork penalties are effective in preventing the selection of segments that share hits.

Noise-free validation. The model optimized for the noise-free scenario achieves strong performance on clean data: a segment recall (TPR) of 93.4%, a precision of 94.6%, and a strict track efficiency of 85.3%. These numbers confirm that the Hamiltonian design and simulated annealing procedure are capable of solving the idealized track reconstruction problem to a high degree of accuracy. The remaining errors are concentrated in regions where two trajectories produce hits that are very close to each other on the same detector layer, a point we return to below.

Sensitivity to noise. The most striking feature of Table 1 is the dramatic performance collapse when the noise-free model is applied to noisy data: the strict track efficiency drops from 85.3% to 1.6%, and the fake rate rises from 5.3% to 53.5%. This is not surprising in retrospect. The noise-free model was tuned with tight geometric tolerances, small angle thresholds, strict curvature consistency, that are well suited to clean data but become overly restrictive when hits are perturbed by Gaussian noise and contaminated by spurious hits. In this regime, true segments can fall outside the acceptance window, while noise-generated segments that happen to satisfy the tight criteria are selected instead, leading to nearly complete reconstruction failure (see Figure 3).

The energy trace of the annealing run confirms this picture (Figure 4). The Hamiltonian fluctuates heavily throughout the run and converges to a minimum of $H = -27.0$ with only 53 selected

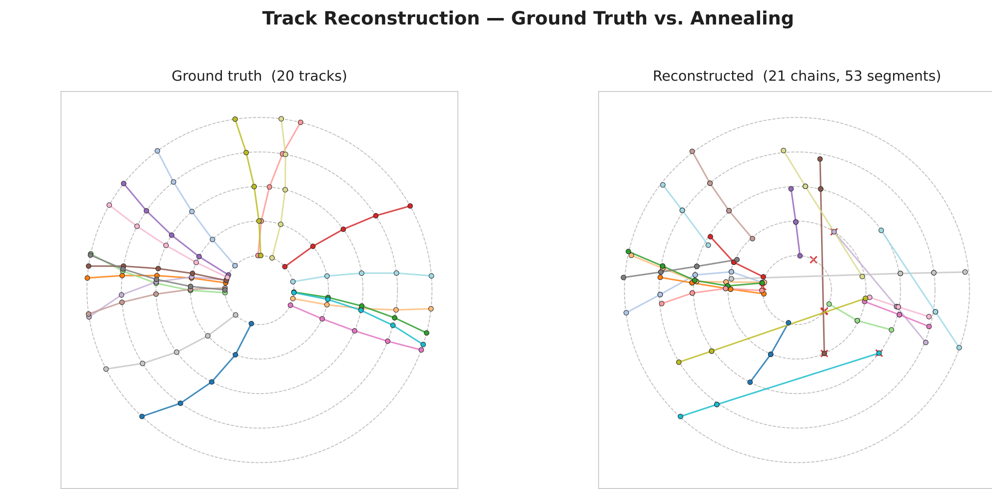

Figure 3: Example of a noisy event reconstructed with the noise-free-optimized model. Left: ground truth tracks. Right: reconstructed tracks. The reconstruction fails entirely (strict track efficiency of 0% for this event), as the model's tight tolerances reject most true segments while accepting noise-induced ones.

segments, far fewer than the 80 true segments expected from 20 tracks across 4 layer gaps. By contrast, the same model applied to a noise-free event (not shown) converges smoothly to $H = -38.5$ with exactly 80 selected segments. The shallower minimum and the persistent high-amplitude fluctuations visible in Figure 4 are characteristic of a rugged energy landscape with many competing local minima of similar depth, which prevents the linear cooling schedule from guiding the system toward the global minimum.

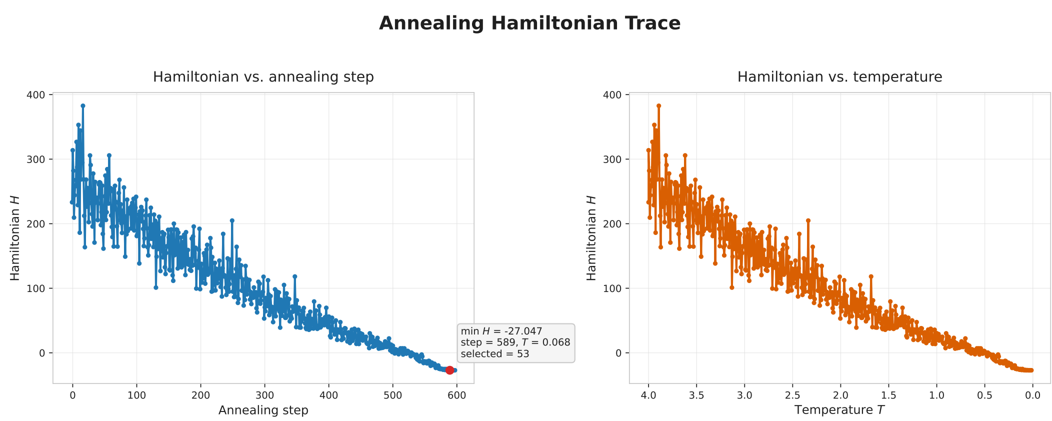

Figure 4: Hamiltonian energy trace for a noisy event reconstructed with the noise-free-optimized model. Left: energy vs. annealing step. Right: energy vs. temperature. The system reaches a minimum of $H = -27.0$ with 53 selected segments, well above the expected 80, indicating that the annealer is trapped in a shallow local minimum.

Noisy-optimized model. Re-tuning the hyperparameters on noisy events yields a model with qualitatively different behavior. On noisy data, the strict track efficiency improves from 1.6% to 30.7%, and the soft track efficiency reaches 58.6%, indicating that the majority of tracks are reconstructed with at most one incorrect segment. Notably, however, this model performs worse than the noise-free model even on clean data (strict efficiency 63.1% vs. 85.3%), which shows that the two regimes require genuinely different trade-offs in the penalty weights: the noisy model must relax geometric tolerances to accommodate measurement uncertainty, but this relaxation also admits more false segments in the clean setting. In other words, the noisy track reconstruction problem is not a simple extension of the noise-free one; the two require models with different effective physics.

Dominant failure mode. After visually inspecting a large number of reconstructed events, we identified a consistent failure mode that accounts for most of the residual errors in the noise-optimized model. When two trajectories produce hits that lie close to each other on the same detector layer, the model can swap the hits between the two tracks without incurring a significant energy penalty, because the resulting segments have nearly the same curvature as the true ones. In energy-landscape terms, the true configuration and the swapped configuration correspond to two local minima with very similar energies, and the annealing procedure has no strong signal to prefer one over the other. This effect is illustrated in Figure 5: the reconstructed tracks appear almost identical to the ground truth at first glance, yet the swap of a few nearby hits causes two tracks to be scored as incorrect under the strict metric.

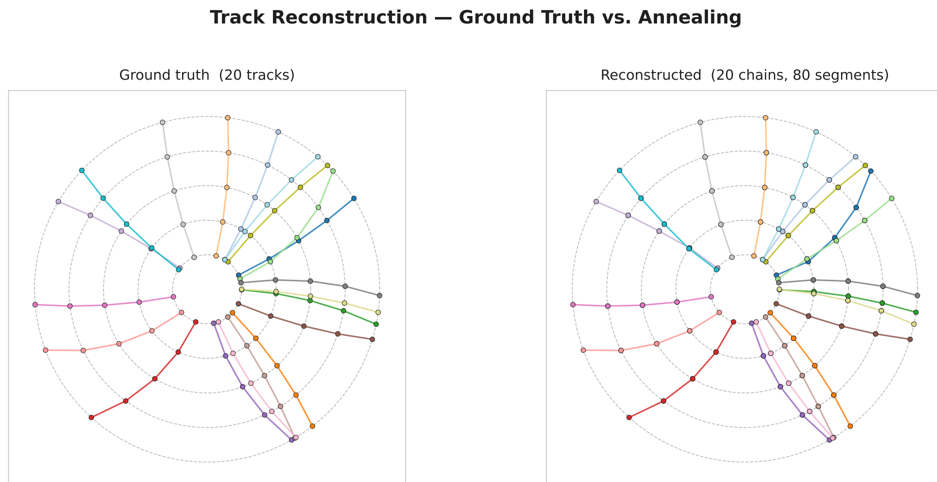

Figure 5: Example of a noise-free event reconstructed with the noise-optimized model. Left: ground truth. Right: reconstruction. The tracks appear visually correct, but hits that lie close together on the same layer are swapped between two trajectories, resulting in a strict track efficiency of only 50%.

This failure mode is exacerbated in the noisy scenario, where spurious hits further increase the density of candidate segments in crowded regions, making it easier for the model to select geometrically plausible but incorrect segments. Importantly, this failure is intrinsic to a model that has successfully captured the large-scale physics of the problem: the errors arise not from a gross misunderstanding of track structure, but from an inability to resolve fine-grained ambiguities between nearly degenerate configurations. The overall topology of the reconstructed event remains correct, as evidenced by the high soft track efficiency (58.6% on noisy data), even when the strict metric penalizes individual segment swaps.

7 Conclusion

We have presented a physics-inspired approach to particle track reconstruction in a simplified 2D detector, formulating the problem as energy minimization over an Ising-like spin-glass Hamiltonian solved via simulated annealing. The Hamiltonian encodes three classes of physical constraints, pairwise segment interactions that penalize bifurcations and reward smooth angles, local field terms that suppress long and non-radial segments, and a three-body curvature term that enforces consistency with circular trajectories in a magnetic field.

On noise-free data, the method achieves a strict track efficiency of 85.3% and a soft efficiency of 93.2%, demonstrating that the Hamiltonian design successfully captures the essential physics of the track reconstruction problem. The inclusion of the three-body interaction term proved critical: without it, preliminary experiments showed frequent selection of segments that do not form smooth trajectories, whereas with it the model can perfectly reconstruct tracks in simple noise-free configurations.

In the presence of measurement noise and spurious hits, however, performance degrades substantially. Re-tuning the hyperparameters on noisy events recovers a strict efficiency of 30.7% and a soft efficiency of 58.6%, a large improvement over the 1.6% strict efficiency obtained by applying the noise-free model directly, but still far from the clean-data performance. The dominant failure mode, swapping of hits between nearby trajectories that produce nearly degenerate energy configurations, reveals a fundamental limitation of the approach: the Hamiltonian relies on local geometric features (angles, lengths, pairwise curvature) that cannot resolve ambiguities when two tracks pass close to each other, because both the true and the swapped configurations satisfy the geometric constraints equally well.

Several directions could address this limitation. First, employing parallel tempering with replica exchange across multiple temperature chains, would improve the ability of the sampler to escape shallow local minima in the rugged noisy energy landscape. Second, running multiple independent annealing restarts and selecting the lowest-energy solution is a simple strategy that directly mitigates the local-minima problem. Third, the Hamiltonian itself could be enriched with longer-range interaction terms, for instance, a global momentum-conservation constraint or a four-body term spanning four consecutive layers, that would break the near-degeneracy between true and swapped configurations by incorporating information beyond the local three-segment window. Finally, extending the model to 3D and scaling it to realistic detector geometries with $\mathcal{O}(10^4)$ hits would be the natural next step toward a practical application, likely requiring a graph neural network or other learned pre-filtering stage to reduce the combinatorial space before annealing.

Despite its limitations in the noisy regime, the approach validates the core premise that track reconstruction can be meaningfully mapped onto a spin-glass optimization problem, and that simulated annealing can navigate the resulting energy landscape to recover physically plausible track configurations. The clear separation between model design (the Hamiltonian) and the solver (annealing) makes the framework modular: improvements to either component can be developed and tested independently, and the Hamiltonian can in principle be solved by alternative methods such as quantum annealing [8] without modification.

Acknowledgments. The C++ source code for the simulation was jointly developed with Nicola Mucciaccio. Giovanni Guidarini implemented the sweep machinery for the Euler cluster, and performed the hyperparameter optimization. Nicola Mucciaccio implemented the plotting code and the evaluation metrics. The results were analyzed jointly by both authors.

REFERENCES

[1] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 2623–2631, 2019.

[2] Sabrina Amrouche, Laurent Basara, Paolo Calafiura, Victor Estrade, Steven Farrell, Diogo R. Ferreira, Liam Finnie, Nicole Finnie, Cécile Germain, Vladimir Vava Gligorov, Tobias Golling, Sergey Gorbunov, Heather Gray, Isabelle Guyon, Mikhail Hushchyn, Vincenzo Innocente, Moritz Kiehn, Edward Moyse, Jean-François Puget, Yuval Reina, David Rousseau, Andreas Salzburger, Andrey Ustyuzhanin, Jean-Roch Vlimant, Johan Sokrates Wind, Trian Xylouris, and Yetkin Yilmaz. *The Tracking Machine Learning Challenge: Accuracy Phase*, pp. 231–264. Springer International Publishing, November 2019. ISBN 9783030291358. doi: 10.1007/978-3-030-29135-8_9. URL `http://dx.doi.org/10.1007/978-3-030-29135-8_9`.

[3] Bruce Denby. Neural networks and cellular automata in experimental high energy physics. *Computer Physics Communications*, 49(3):429–448, 1988. doi: 10.1016/0010-4655(88)90004-5.

[4] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi. Optimization by simulated annealing. *Science*, 220(4598):671–680, 1983. doi: 10.1126/science.220.4598.671. URL `https://www.science.org/doi/abs/10.1126/science.220.4598.671`.

[5] Andreas Krause and Jonas Hübotter. Probabilistic artificial intelligence, 2025. URL `https://arxiv.org/abs/2502.05244`.

[6] Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller. Equation of State Calculations by Fast Computing Machines. *Journal of Chemical Physics*, 21(6):1087–1092, June 1953. doi: 10.1063/1.1699114.

[7] Carsten Peterson. Track finding with neural networks. *Nuclear Instruments & Methods in Physics Research. Section A: Accelerators, Spectrometers, Detectors, and Associated Equipment*, 279(3): 537–545, July 1989. ISSN 0168-9002. doi: 10.1016/0168-9002(89)91300-4.

[8] Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, and Maria Spiropulu. Charged particle tracking with quantum annealing optimization. *Quantum Machine Intelligence*, 3(2):27, 2021. doi: 10.1007/s42484-021-00054-w. URL `https://doi.org/10.1007/s42484-021-00054-w`.

Declaration of Originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies.

☐ I hereby declare that I authored the work in question independently, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies.

☒ I hereby declare that I authored the work in question independently. In doing so I only used the authorised aids, which included suggestions from the supervisor regarding language and content and generative artificial intelligence technologies. The use of the latter and the respective source declarations proceeded in consultation with the supervisor.

Title of paper: Particle Track Reconstruction as a Spin-Glass Optimization Problem

Authored by:

Last name(s): Guidarini
First name(s): Giovanni

With my signature I confirm the following:

- I have adhered to the rules set out in the Citation Guidelines.
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date

Signature